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The impact of socio-demographic factors on occupants' thermal comfort and sensation: An integrated approach using statistical analysis and agent-based modeling

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ABSTRACT

The thermal comfort and thermal sensation of occupants can significantly impact the overall performance of the building. However, very little is known about how occupants perceive or thermally respond to their building interior layouts or what factors affect their thermal comfort and sensation, particularly in low-income residential homes. Through a post-occupancy evaluation of 394 occupants from 190 households in Chittagong, Bangladesh, this study examines the socio-demographic factors (i.e., gender, age, and home ownership) that influence occupants' thermal comfort and sensation. The chi-square, Cramer's V, and Lambda tests were used to statistically analyze the impact of selected socio-demographic factors. The results showed that female occupants were significantly less satisfied than male occupants with their existing thermal environment. However, the study did not find any influence of age or home ownership on occupant thermal comfort. Furthermore, comprehensive agent-based modeling (ABM) was implemented to support the results of the statistical analysis of the influence of gender on occupants' thermal comfort levels. The ABM was designed based on the theory of planned behavior. As a simulation output, the ABM model stochastically estimated occupants' thermal comfort and sensation level in terms of the predicted mean vote. Similar to the statistical analyses, the results of the ABM also indicated better thermal comfort and thermal sensation for male occupants than for female occupants. The results of the study can serve as a guideline for better understanding the socio-demographic factors involved in the modeling or remodeling of occupant thermal comfort in developing economies.

1. Introduction

Indeed, energy efficiency and thermal comfort are two significant driving forces in building design and operation [1]. It is estimated that heating, ventilation, and air conditioning (HVAC) systems account for approximately 55 % of building operational energy consumption [2]. The primary purpose of HVAC systems is to meet occupants' thermal comfort requirements by regulating indoor temperature, humidity, and air quality. Building designers often simulate thermal comfort requirement during the design phase based on assumptions about how occupants will perceive within the indoor environments [3]. However, it is true that occupant's perception, including their response to thermal comfort, can vary and may go beyond designer's initial expectations. This discrepancy between actual and anticipated response levels can impact thermal comfort in real-world scenarios. For instance, several studies [1,4] have revealed significant deviations between the predicted mean vote (PMV) model and the actual thermal comfort experienced by occupants in real-world field studies. In addition, research has demonstrated that the predicted neutral temperature of the Fanger model [5] can deviate significantly from the actual thermal sensation reported by occupants. These deviations can exceed 3.6 °C and are highly dependent on individual perception [1]. Thus, both architectural and non-architectural factors (e.g., psychological, social, etc.) play significant roles in influencing occupant comfort levels within indoor environments [6]. Moreover, occupant spend more than 90 % of their time in an indoor environment [2]. So, better indoor environment design not only reduces energy consumption but also improves occupant working efficiency and well-being. In addition, occupants' thermal preference

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differs due to various socio-demographic factors, such as age, gender, home ownership, and body mass index (BMI) [7,8]. Therefore, it is crucial to investigate the influence of architectural and socio-demographic factors on occupants' thermal comfort preferences in order to provide a more comfortable environment [2].

From the architectural and socio-demographic perspectives, some studies [6,9] have found that office interior layout configurations have a prominent impact on occupants' comfort level in both summer and winter. Another study [10] also revealed a strong correlation between interior layout and occupant thermal comfort. Herein interior layout refers to the spatial arrangement and configuration of indoor spaces within a building, including the room orientation, arrangement, corridors, partitions, furniture, and other elements that shape the overall layout and organization of the indoor environment. However, the spatial impacts of architectural components on occupants' thermal comfort, specifically regarding a particular building interior layout in low-income contexts, have not been fully investigated, and the explicit representation of these impacts remains unclear [2]. In addition, because of the difficulty of quantifying and observing them, individual factors such as psychological and socio-demographic variables have rarely been assessed in thermal comfort modeling [11,12]. Estimating personal comfort while considering these individual factors in a dynamic environment is a challenging but essential task for optimizing overall building performance. Moreover, improving occupants' well-being, as occupants' personal comfort may be highly dependent on their individual differences, such as age, gender, and body mass index [12]. Hence, to assess the influence of socio-demographic factors on occupants' thermal comfort and sensation level within the particular interior layout context, multiple hypotheses should be proposed and tested based on a wide knowledge of occupant perceptions/comforts that might be subject to change or transformation [13–15]. Fewer studies have demonstrated the relationship between socio-demographic factors and the thermal comfort levels experienced by various occupant groups. These factors do indeed influence how occupants respond to and behave in their surrounding built environment [12,16]. Recently, a growing body of research approaches and models have investigated several factors, especially socio-demographic factors, that contribute to occupants' thermal comfort in a holistic way [1,3,17].

In traditional thermal comfort modeling approaches [1,6,18], researchers typically follow a single methodology or process rather than a combination. Four main methodologies have been implemented: probabilistic or stochastic analysis, data mining, statistical techniques, and agent-based modeling (ABM) [19,20]. The first three are well-established approaches for thermal comfort and energy consumption analysis, and they have continued to predominate in the most recent research [21]. However, few studies have analyzed the influence of socio-demographic factors on occupants' thermal perception/comfort by integrating statistical and ABM approaches [22]. Thus, statistical analysis and ABM were selected as the methodological approach in this study based on several considerations. Statistical analysis enables the construction of a statistical relationship between occupants' thermal comfort/perception and relevant socio-demographic factors such as gender, age, and home ownership. ABM has a simulation-based modeling structure that comprises single or multiple autonomous actors called "agents," who interact with each other and their surrounding built environment according to specified rules [23]. These rules are fundamental during the thermal comfort simulation process, as they allow accurate characterization of how and when agents, by analogy with actual humans, interact or cooperate by following conditions within their environments. Thus, this study included a statistical analysis followed by an ABM approach to ensure comprehensive analysis and a flexible modeling environment. Several other studies have similarly revealed that in-situ occupancy data (e.g., from surveys, monitoring) can be combined with the simulation approach to help stakeholders understand occupants' thermal comfort levels [24].

In this research, a questionnaire survey was conducted to explore the impact of the gender, age, and home ownership of occupants on their thermal comfort and thermal sensation in their existing interior layouts, covering 190 households in a low-income economy (Chittagong, Bangladesh). During the questionnaire survey, parameters related to interior layouts, such as orientations, areas of floors, and walls are considered consistently. Moreover, the environmental parameters has also been kept similar across the selected households to facilitate a better understanding of the influences of specific socio-demographic factors. The collected data were stratified based on gender, age, and home ownership (i.e., owner or tenant) to understand the impact of these socio-demographic factors. A 7-point Likert scale was used to measure the occupants' thermal comfort level within the existing interior layouts. As an additional parameter, thermal sensation levels in terms of predicted mean vote (PMV) indices were also collected during the questionnaire survey. Both thermal comfort and thermal sensation were measured on a 7-point Likert scale. Thermal comfort qualitatively describes the pleasantness of a thermal stimulus (like or dislike), whereas thermal sensation describes the magnitude of the thermal experience (warm or cool). A notable challenge during the most crucial part of occupants' thermal comfort/perception studies is reliability analysis. However, this has rarely been performed to support the study data in prior research. Therefore, this research also aimed to fill this gap by expanding the results of the statistical analysis through constructing an ABM model and verifying the statistical performance to obtain more realistic PMV indices. Recently, ABM has been one of the most popular approaches for analysis of occupant stochastic perceptions and thermal comfort, and has been recommended by several studies [20,25]. The results facilitate mapping of the interaction between occupants' thermal response and socio-demographic factors within the specific building interior layout, which can be used for both theory and model construction [2].

The rest of this article is structured as follows: Section 2 presents the literature review, Section 3 describes the research method, Section 4 presents the results and discussions, and Section 5 concludes the study.

2. Literature review

Several researchers have observed a strong correlation between occupants' perception of control over their indoor environment and productivity. In several countries, researchers have concluded that room size/layout plays a vital role in the thermal comfort of aging houses [26–29]. Investigations of thermal comfort management in numerous regions also indicate that per-capita energy consumption rises with the deterioration of houses [28,30,31]. A survey study revealed that occupants living in naturally ventilated buildings in Singapore were exposed to higher operative temperatures than the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) comfort standard for naturally conditioned spaces [32]. However, occupants could still improve their comfort level by turning on mechanical fans and using windows and doors for cross-ventilation. Nonetheless, that study did not consider the gender, age, or other demographic information of occupants while assessing the occupants' thermal comfort level. In Germany and other European countries [30,33], researchers have observed that monthly energy consumption rises with the age of occupants, especially energy consumption for maintaining thermal comfort and hot water production. As more than 90 % of the aging houses in Germany host two-person or single-person households, a loss of economies of scale is presumed to be a cause of their higher energy consumption for better comfort management [34–36]. In support of thermal comfort, Azhar and Pisello [37,38] presented a target direction for planning interventions (peer to peer communication). This was based on the persuasive systems design process [39], which can aid the improvement of effective efforts to change individuals' behavior and enhance sustainability. This information, alongside general behavior transformation approaches (e.g., windows on/off, interior layout adjustment), design with target tools (e.g., intervention for comfort management), and planning for energy

Table 1
Quantitative studies of occupants' thermal comfort and behavior.

Building type	Region	Factors						Methods/Approaches	Data sources	References
		Gender	Age	Ownership	Income	IEQ	Layout			
Residential	Luxembourg	○	○			○		•Statistical analysis[a]	•Survey •Sensor	[54]
Student Hall	USA	○				○		•Chi-square test	•Survey	[55]
University Dormitory	Singapore		○			○		•Welch ANOVA •Games–Howell post hoc test	•Survey •HOBO data loggers	[32]
Residential	Hong Kong	○	○					•Chi-square test •Kruskal–Wallis test •Mann–Whitney *U* test	•Survey	[56]
Residential	Finland	○	○	○	○	○		•Logistic regression	•Survey •CalcTM Indoor Air Quality Meter 7525	[52]
Office	USA	○	○			○		•Whitney *U* test •Kruskal–Wallis test	•Survey	[53]
Office	India	○						•Logistic regression •*t*-test	•Survey •Thermo-hygrometer	[7]
Office	Bahrain	○	○			○	○	•One way ANOVA •*t*-test	•Survey	[44]
Office	European countries	○	○			○		•Logistic regression	• Survey	[57]
Office	Australia					○	○	• Regression • Two-way ANOVA	• Survey	[45]
Office	USA	○				○	○	• Logistic regression • Chi-Square test	• Survey	[50]
Office	USA	○	○			○	○	• Logistic regression • *t*-test	• Survey • Data loggers	[58]
Residential	Italy	○		○				• Multivariate regression	• Online questionnaire	[59]
Nursing Home	China	○	○					• ANOVA	• Survey • Sensor network	[60]
Educational	Egypt					○		• Statistical analysis[a]	• Survey • Interviews	[61]
Office	Austria					○	○	• Agent-Based Modeling	• Survey • Modcam sensor • CCTV camera footage	[49]
University	China	○				○		• Kruskal–Wallis test • Pearson correlation	• Survey • Sensors	[62]
University	Malaysia	○				○	○	• ANOVA • *t*-test	• Survey • Sensors	[63]
Residential	China		○		○			• Regression	• Survey	[64]
Residential	India		○		○			• Chi-square test • ANOVA	• Survey • Indian database for construction materials	[16]
Residential	Denmark	○	○		○	○		• Regression	• Survey	[65]

[a] The specific statistical analysis method has not been mentioned in the study.

efficient behaviors without compromising thermal comfort, was applied to improve the interventions presented [22,38,40]. Other studies have demonstrated a substantial gap between energy-saving prospects and the availability of occupants' thermal comfort data to aid interior design in the early stage [40,41]. Delzendeh [22] indicated that interior layout could influence an occupant's movement and presence in a particular area because it can affect the occupant's conduct, thermal comfort, or activities within an indoor space. Thus, the occupant's probability of movement and presence in a particular area is based on several spatial parameters (e.g., distance from switch locations, window/blinds, thermostat etc.) that can be simulated. The authors of the present study have conducted a comprehensive literature review (see Table 1) that includes the latest studies on the relationship between occupant perception and thermal comfort. The review incorporates the latest relevant articles and includes information on building types, region, factors considered, tools/platforms, and data sources of occupant behavior. Overall, the studies reveal that several factors indicated in Table 1 may considerably affect occupants' thermal comfort and sensation. In general, the most significant factors are spatial, environmental, and building-related factors, along with a few auxiliary parameters (e.g., gender, age). For instance, spatial factors such as occupants' actions and movement rules [36,42–45], windows and blinds [46–48], interior layouts [45,49,50], and intervention context [37,40,51] are emphasized over socio-demographic factors. Moreover, a few studies [52,53] have investigated the influence of a single socio-demographic factor (e.g., gender, home ownership) in high-income rather than low-income backgrounds. Above studies, however, did not consider the individual thermal sensation levels (i.e., cold or hot) of occupants in relation to the particular socio-demographic factor. Thus, it is important to confine the spatial plan according to a range of parameters, including both indoor environmental quality (IEQ) and non-IEQ factors, to completely determine its impact on occupants' thermal comfort and sensation level, not only in high-income but also in low-income cultural backgrounds [45,48,54]. This comprehensive study aimed to elucidate the impact of socio-demographic factors (i.e., gender, age, and home ownership) on the thermal comfort and sensation levels of occupants in the low-income context.

3. Research methods

In this study, a questionnaire survey was distributed to households from a selected location to explore how socio-demographic factors affect their thermal comfort and sensation levels within their existing interior layout. Valid data from the respondents were statistically analyzed and extended ABM was conducted to verify the statistical reliability. The overall research methodology is shown in Fig. 1.

3.1. Target selection

Covering an area of 2510 km^2 and located 29 m above mean sea level, Chittagong is a large port city on the southeastern coast of Bangladesh with a population of 2.59 million. The rainy summer season lasts from May to October and is warm, wet, and cloudy. This survey study was performed in the summer months of June and July 2021. Households from 19 residential buildings in central Chittagong were considered, including tenants and property landlords. The surveyed buildings were generally 3–4 stories high and 10–30 years old.

3.2. Data collection for statistical analysis

In this study, 394 occupants were recruited from the selected locations for the semi-structured questionnaire survey. All of the occupants were aged between 17 and 75. As suitable days for occupants, the data collection period was determined from June 1 to July 31, 2021. The selected respondents were supplied with a data sheet clarifying the study aim and objective. At the same time, the occupants' consent was

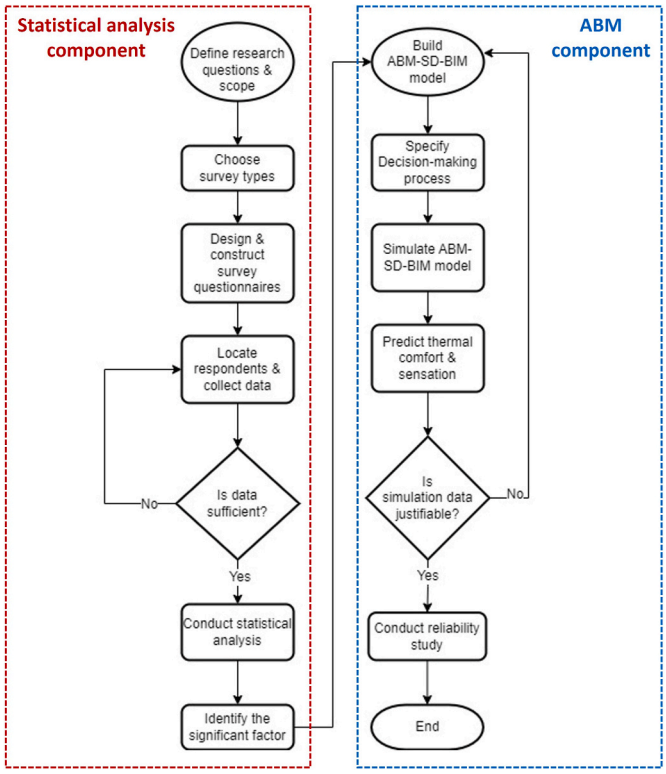


Fig. 1. Research framework.

acquired using a standard consent form. Next, the respondents were asked to provide their basic information and their thermal perceptions within their interior layouts by answering the survey questions. The questionnaire survey form consisted of two main sections. The first section included items on the occupants' socio-demographic details (e.g., gender, age, education level, home ownership). The second section comprised items on the occupants' thermal perception in terms of thermal comfort and sensation scores (see Appendix I). In this section, the occupants were asked to report their thermal comfort and sensation levels (i.e., PMV indices) on a 7-point Likert scale based on the ASHRAE standard (see Table 2). Environmental data (e.g., temperature, humidity, CO_2 level) were monitored using smart meters (3 in 1 carbon dioxide detector with wireless air quality monitor) and a local weather website (https://live.bmd.gov.bd/). Furthermore, manually measured building physical characteristics such as spatial configuration, floor area, and building orientation were kept constant across the study period. More elaborately, the selected interior layouts were chosen within similar geometric dimensions, floor levels, orientations to maintain the consistent profile. These building physical characteristics, such as spatial configuration (e.g., size, shape, density, etc.), floor area (e.g., 180–200 sqft), and building orientation (e.g., south-east), play a crucial role in the subsequent ABM-SD-BIM simulation as input data. These measurements provide essential information for accurately representing the physical attributes of the building within the simulation platform. Moreover, to ensure consistency in various environmental parameters across different households with similar layouts and orientations, data collection time slots were carefully chosen to prioritize data consistency and homogeneity. For instance, data collection was scheduled from 12:00 p.m. to 3:00 p.m., corresponding to the peak temperature period during a typical sunny day. As a result, all households maintained similar outdoor and indoor air temperature as well as CO_2 levels.

The data samples were gathered based on their gender, age, and home ownership. Later, a descriptive statistical evaluation was accomplished that quantitatively evaluated the data gathered from the survey questionnaire. Data samples were studied using the Statistical Package

Table 2
Scales of thermal comfort and sensation.

Thermal votes	−3	−2	−1	0	+1	+2	+3
Thermal comfort	Strongly dissatisfied	Dissatisfied	Slightly dissatisfied	Neutral	Slightly satisfied	Satisfied	Strongly satisfied
Thermal sensation	Cold	Cool	Slightly cool	Neutral	Slightly warm	Warm	Hot

for the Social Sciences (SPSS), version 22 for Windows. The chi-square, Cramer's V, and Lambda tests were used to examine the influence of socio-demographic factors on the occupants' thermal comfort and sensation. To calculate the differences between categorical occupant groups, the chi-square test for independence with $\alpha = 0.05$ was used to assess whether occupants' thermal comfort was related to gender, age, and home ownership. Additionally, the Cramer's V association test (symmetric measures) was used to quantify the strength of the associations between the occupant classes or groups, and the Lambda correlation test (directional measures) was used to identify the direction of the associations. The interpretations of the Cramer's V and Lambda values are shown in Table 3 [66–68].

3.3. Model development for ABM

In this study, a comprehensive ABM model was developed, and a reliability analysis was performed to test and verify the statistical performance, followed by a comprehensive discussion about the study. In this context, this research suggests an ABM model to simulate occupancy profiles and interactions within the indoor environment that can incorporate both architectural and socio-demographic factors. By considering a similar approach to empirical ABM models implemented in social simulation studies, the proposed ABM model also incorporates statistical information to calibrate the model. Furthermore, the construction of the hybrid model involved utilizing general occupancy profiles obtained from the questionnaires survey. When simulating the interactions within the ABM platform, the model included gender, age, and home ownership. Notably, the model predominantly considered the particular socio-demographic factors shown to be statistically significant parameters by the previous SPSS analysis. The remaining parameters were considered as auxiliary factors during the ABM model construction. The key outcomes of the ABM model were occupancy patterns based on socio-demographic factors as well as the thermal comfort and sensation levels of occupants within a consistent interior layout. The approach of analyzing the statistical and ABM model data helps to understand the population sub-groups and diverse behavior patterns within the heterogeneous agent population. This also revealed that overall building performance is highly dynamic. In addition to socio-demographic factors, building performance relies on various other parameters, such as climate, indoor environment, number of people, clothing level, heat gain, and design features such as layout design, placement of furniture, and orientation [49,69]. These strongly influence the building's indoor performance during the operation stage. Thus, an integrated methodology is needed to more accurately simulate the influences on stochastic occupants' thermal comfort and sensation within the particular layout [23]. Moreover, besides the fundamental ABM model, it is necessary to integrate other supporting modeling approaches such as system dynamics (SD) and building information modeling (BIM) that incorporate additional essential parameters. Hence, in this study, a comprehensive thermal comfort modeling structure was established that represented a hybrid approach using ABM, SD, and BIM (Fig. 2).

Table 3
Interpretation of Cramer's V and Lambda.

Interpretation	Cramer's V	Lambda
Very strong	>0.25	+1.00
Strong	>0.15	+.30–.99
Moderate	>0.10	+.10–.29
Weak	>0.05	+.01–.09
No association	>0	0

Note: $p < 0.05$ was considered to indicate statistical significance.

Thus, the modeling approach supports message/data exchange between the ABM, SD, and BIM to fully capture the diverse elements in the occupancy-based thermal comfort analysis of a typical interior layout. Here, the application programming interface (API) Dynamo-API was implemented for data exchange between the BIM model and the ABM–SD model. Dynamo-API is a graphical programming tool that efficiently collaborates with the BIM model by offering access to the Revit API in an accessible way. It also permits new users to create specified programs by building visual interaction between several components within a hybrid framework system. The detailed process of model integration is described in previous research [23,69]. Herein, the

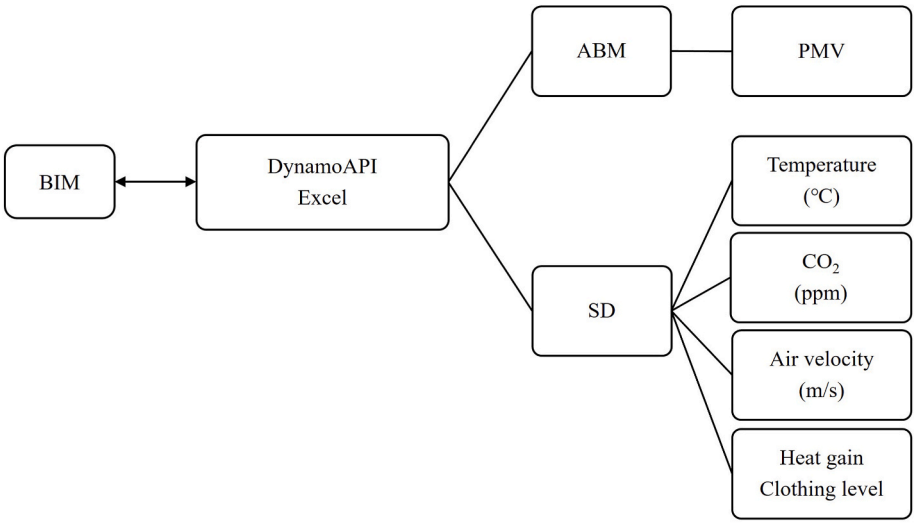


Fig. 2. ABM–SD–BIM framework.

ABM–SD was implemented using the AnyLogic modeling tool, a broadly developed simulation environment in the business, sociology, and engineering domains. The construction of the model is described in the following section.

3.3.1. Rules and segments of the proposed model

As described in the literature [20,23,25,69], the major elements of an ABM model comprise a set of agents, a particular topology, and rules that refer how the approach drives within the environment. In relation to this modeling context and background, the usual elements of an ABM structure are illustrated in Fig. 3. The proposed ABM approach is depicted with respect to its three mains parts in Fig. 4.

In part 1, the agents in the model are occupants, specifically the building occupants. Occupants are complex agents in terms of their attributes and characteristics. It is challenging and often avoidable or highly difficult to catch all of the occupants' characteristics or attributes. In this paper, the model considered the occupants' attitudes, subjective norms, and perceived behavior control according to the theory of planned behavior (TPB) [69,70]. The TPB is typically considered an advanced version of the theory of reasoned action (ToRA) with an additional behavior control parameter, "perceived behavior control," that is suitable for analyzing individual behavioral intention rather than other available modeling approaches (ToRA, Belief–Desire–Intention (BDI) etc.). Therefore, in this study, the TPB was considered as a decision-making process within the ABM approach. Other psychological (e.g., rebound-effect, perception on layout preferences) and non-psychological parameters (e.g., physical characteristics) were also incorporated as supporting and influential factors during the modeling process.

In part 2, the environment in which agents interact within the model is appropriately described (e.g., SD and interior layout components in the Revit model). Here, the other building indoor conditions (e.g., building physical characteristics) and the built environment (e.g., air circulation, temperature, humidity) states were considered. Specifically, these data are primarily used and exchanged through the BIM Model (Revit) and System Dynamics components using the DynamoAPI, as mentioned in Appendix-II. The ambient environment is the direct driver or stimulus that controls the agents' thermal comfort/sensation values, which are also linked to the SD model.

In this context, building physical characteristics primarily play a supporting role during the simulation process. The data related to building physical characteristics, which were collected during the survey period, are crucial for creating an artificial environment within the simulation platform, the ABM-SD-BIM. These attributes are essential in accurately representing the built environment within the simulation, thereby enhancing the realism and effectiveness of the analysis. By incorporating these physical characteristics, the simulation can capture the influence of the building's spatial configuration, floor area, and orientation on thermal comfort and sensation experienced by occupants.

In part 3, the agent interaction and learning process was initiated upon the completion of the decision-making process. If the indoor conditions are outside of the acceptable range for the agent, it is expected that the agent will experience discomfort, and comfort otherwise. However, occupant agents can adjust the state of building indoor components (or positive perception of layout) to reach a higher comfort level. The ambient indoor environment is not the only external feature that influences occupants' behavior in reality [19,69]. For instance, gender, age, economic circumstances, and other social or individual

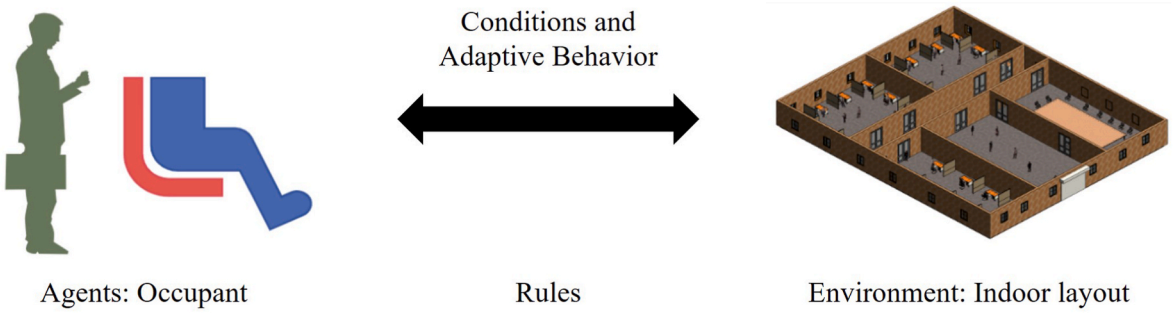


Fig. 3. Proposed ABM structure within the building indoor environment.

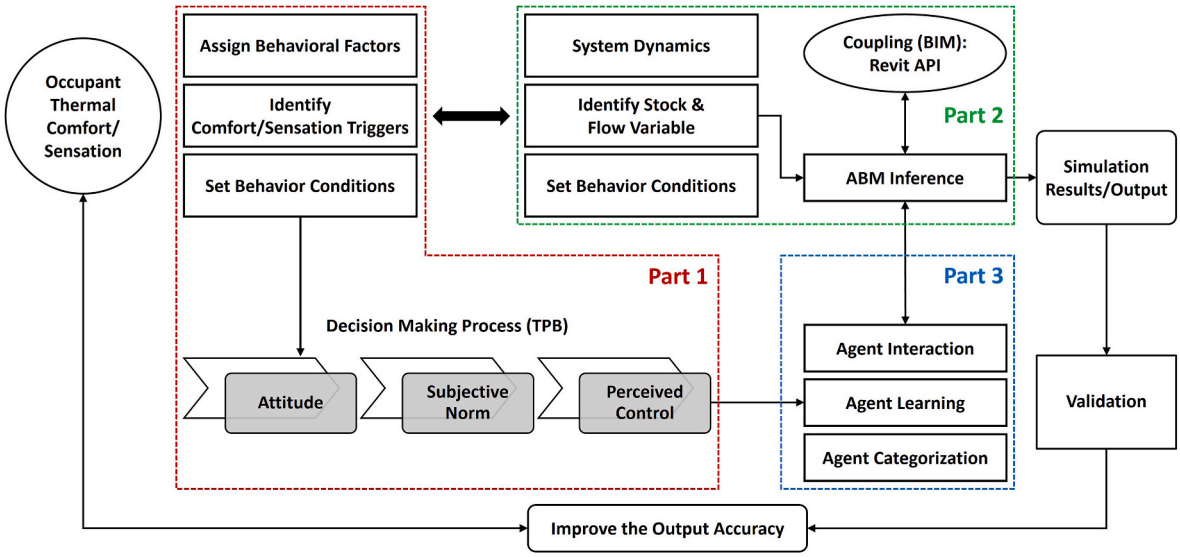


Fig. 4. Three main parts of the ABM framework.

factors might also affect occupants' thermal comfort profile, especially for residential inhabitants. Thus, this model also considered the occupants' gender, age, and home ownership, as these factors had also been tested statistically in the previous step. In the following step, occupant agents were classified/categorized based on the socio-demographic rules incorporated within the model. Ultimately, the proposed ABM model estimated the occupant individual thermal comfort levels based on their socio-demographic factors. Thus, the proposed ABM model considered specific parameters within the interior layouts. In this model, the primary determinant of occupants' thermal perception and comfort is the indoor environmental stimuli, as opposed to other social factors. This suggests that the ABM model focuses on understanding how the indoor environment, including factors like temperature, humidity, and air quality, influences occupants' thermal experiences, giving more weight to environmental stimuli over social factors [22]. The structured state charts for the agent-specific tasks, relationships, and purposes are explained in the following sub-sections and depicted in Fig. 5.

3.3.1.1. Decision-making process (DMP) of occupant agents. The TPB was implemented to describe the decision-making process for the occupant agents within the ABM platform. The TPB implemented within the state chart in Fig. 5 (a) was extended to the "Attitude, Subjective norm, Perceived behavior control, and Categorization" loop for further identification. A comprehensive explanation of the loop is as follows: (i) Attitude: An occupant agent considers its belief based on the surrounding environment. That is, an occupant's beliefs are influenced by the environment in which they are situated; (ii) Subjective norms consist of S_Norm and O_Norm: An occupant agent adapts and devotes some time to evaluate its self-perception (e.g., S_Norm) as well as the perceptions of others (e.g., O_Norm); (iii) Perceived Behavior control: Based on its surrounding environment as well as occupants' attitudes and subjective norms, an occupant agent control its positive (+1) or negative (−1) perceptions/decisions; and (iv) Categorization: An occupant agent is categorized according to socio-demographic groups and proceeds to the next state chart in Fig. 5 (b). A sample of the coding for some specific state chart arrangements within the AnyLogic platform is shown in Appendix III.

3.3.1.2. Occupants' thermal comfort prediction based on socio-demographic factors. Upon completion of the decision-making process, occupant agents were divided into three socio-demographic groups (i.e., gender, age, and home ownership). The maximum number of occupants from each group was also assigned using the probability functions and the data collected from the field survey. At any given time, the model can identify the gender, age, and home ownership of each occupant agent through a triangular distribution, which is implemented as a functional form of areas for fuzzy logic because of the straightforward applicability of fuzzy logic to the model building process. Initially, as a default, the occupants' thermal comfort/satisfaction level with the given environment was assigned a value of 0 on the 7-point Likert scale for each group. The Likert scale value was calculated in each run and ranged from −3 (strongly dissatisfied) to +3 (strongly satisfied). As a dynamic simulation process, most of the dynamic parameters and variables were considered, whereas the event parameters were mainly counted as model actions in a cyclic mode. For the layout-related parameters and variable selection, the results of Uddin et al. [69] were also taken into consideration.

3.3.1.3. Comprehensive thermal sensation prediction. The last state chart (Fig. 5(c)) was used to calculate occupants' thermal sensations subject to the PMV indices. The model was able to calculate the thermal sensation values for each gender, age, and ownership group equally. However, this state chart only counted the most influential socio-demographic factors generated from the previous state. For instance, if gender represents the

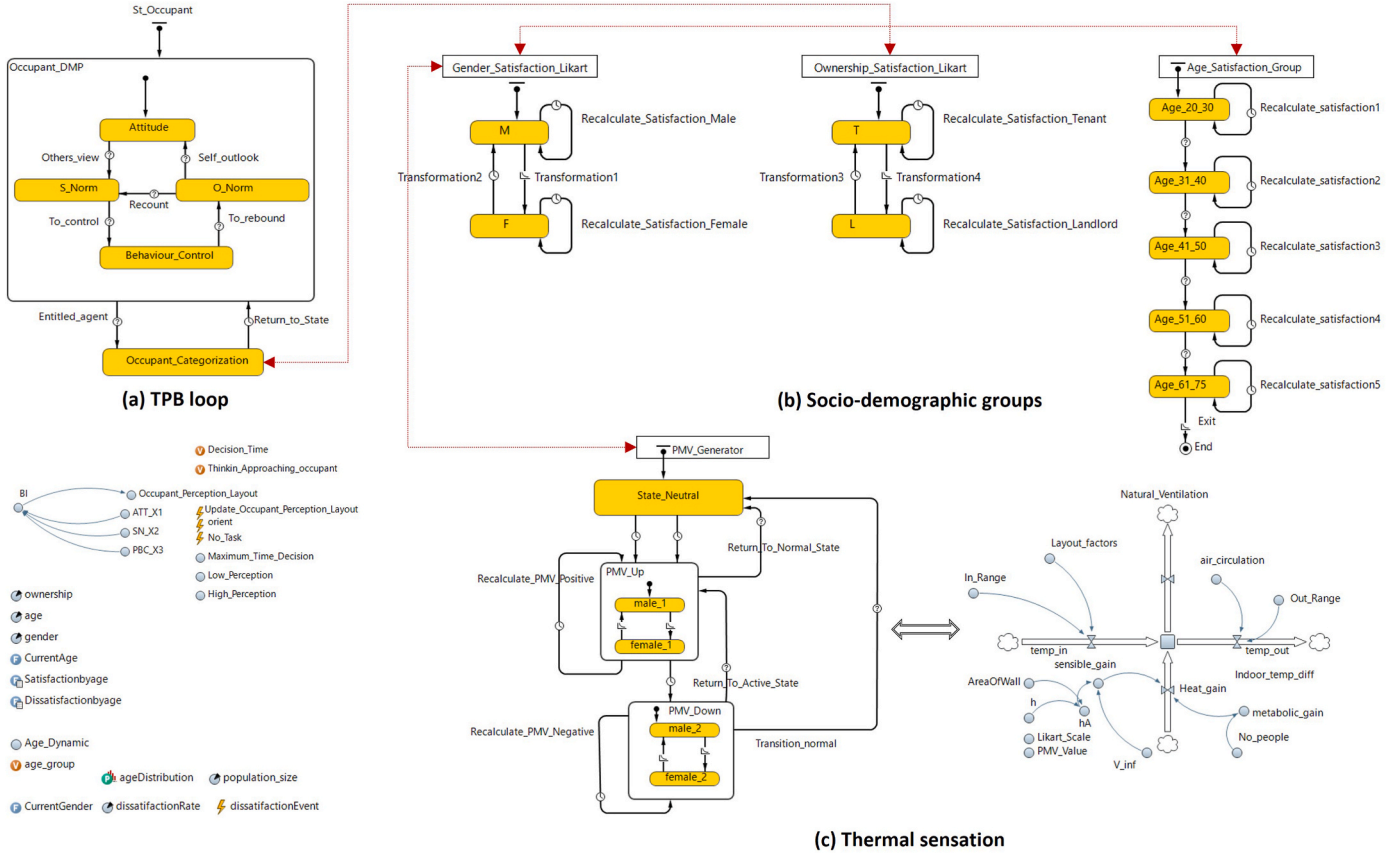


Fig. 5. Multi-structured state charts for ABM.

statistically significant factor, then state chart's first component (i.e., State_Neutral) represents the gender-neutral state where the initial PMV value is zero. The next component (i.e., PMV_Up) characterizes the positive PMV zones for both male and female occupants. The required trigger parameters for detecting thermal sensation values are represented using the system-dynamic components. During the simulation run, the occupant agent moved to the last zone (i.e., PMV_Down) if the comfort level went down or indicated negative PMV values (e.g., PMV = −1). Thus, the total numbers of male and female occupants, including their thermal sensation and thermal comfort level, are counted within each zone. The recalculate transition was used to capture the precise values during the simulation process. A similar process is also applicable for the other two socio-demographic factors.

4. Results and discussion

4.1. Statistical analysis

From the initial statistical results, males and females represented 55 % and 45 % of the occupants, respectively (Fig. 6(a)). There were five age groups (Fig. 6(b)) in the data sample: approximately 24 % of occupants belonged to the age group 20–30, 17 % of occupants were aged 31–40, 24 % of occupants were aged 41–50, 18 % of occupants were aged 51–60, and 17 % of occupants were aged 61 and beyond. Meanwhile, valid responses from homeowners or landlords formed about 51 % of the responses, and tenants comprised the remaining 49 % (Fig. 6 (c)). Table 4 demonstrates the influence of socio-demographic factors on the occupants' thermal comfort as determined by the chi-square, Cramer's V, and Lambda tests.

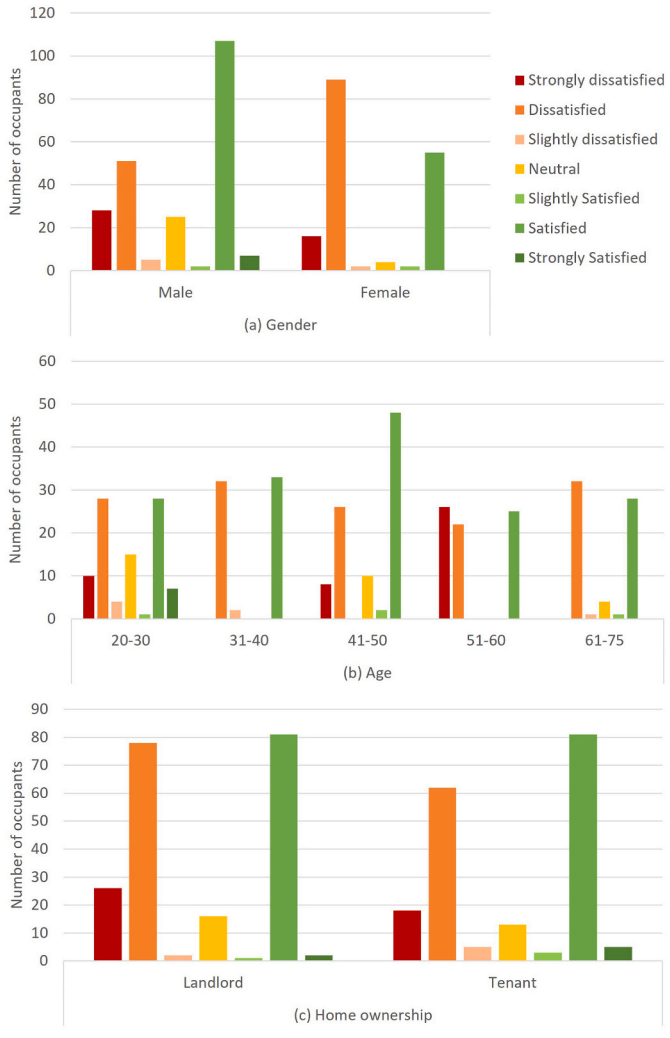


Fig. 6. Occupants' thermal comfort in a given interior layout by (a) gender, (b) age, and (c) home ownership.

The chi-square test indicated a statistically significant effect of gender, χ^2 (1, N = 394) = 13.71, p = 0.001. Additionally, both the Cramer's V (0.430) and Lambda (0.353) parameters indicated a strong association between occupant gender and thermal comfort within a given interior layout.

However, the chi-square test showed statistically non-significant findings for home ownership category (p = 0.647) and age group (p = 0.101). For the home ownership category, the Cramer's V and Lambda values were 0.108 and 0.056, correspondingly, indicating moderate and weak relations between home ownership category and thermal comfort. For age group, the Cramer's V and Lambda values were 0.297 and 0.130, respectively, denoting strong and moderate relations between age and thermal comfort.

From the above statistical review, the research outcomes exposed a significant impact of occupant gender on the thermal comfort. The females were more dissatisfied with the thermal comfort than were the male occupants. No influence of ownership type or age group on thermal comfort was found. Homeowners and tenants showed similar thermal comfort levels within the interior layout. Older occupants (aged 61–75) were noticed to be more thermally satisfied than other age groups. This was also weakly supported by the Cramer's V and Lambda scores for occupant age groups (e.g., strong and moderate relationships). Previous studies [50,71] have also showed that female occupants were less satisfied than male occupants, and the eldest age group had a substantial gender influence. Another study [72] also indicated that the middle age group (i.e., 35–54 years) had a lower thermal comfort level for all IEQ factors, while the older age group (>60 years) exhibited a higher thermal comfort level.

4.2. ABM model output

After multiple simulation runs in the ABM platform, the individual comfort pattern was categorized in terms of male and female occupants, as only gender plays an important role, as revealed by the earlier statistical findings. The model could also generate thermal comfort data based on the other two socio-demographic factors (i.e., age, and ownership); however, only the gender-based PMV index profiles were included, as the other two socio-demographic factors did not display any significant difference between groups. Therefore, in terms of the ABM simulation's final output, the following sections discuss the PMV index profiles for male and female occupants. The simulation results showed that the individual PMV index profiles for a specific period were entirely random, as shown in Fig. 7.

The model produced an individual PMV index profile during the simulation phase, which indicated that the thermal comfort level for both male and female was not neutral (PMV = 0). By gender, the male occupants perceived a slightly warm environment (PMV >0), whereas

Table 4
Impact of socio-demographic factors on thermal comfort.

Socio-demographic factor	Group/ Class	Chi-square test (Pearson chi-square)	Cramer's V test (symmetric measures)	Lambda (directional measures)
Gender	Male Female	**0.001**	0.430	0.353
Age group	20–30 31–40 41–50 51–60 61–75	**0.101**	0.297	0.130
Home ownership	Landlord Tenant	**0.647**	0.108	0.056

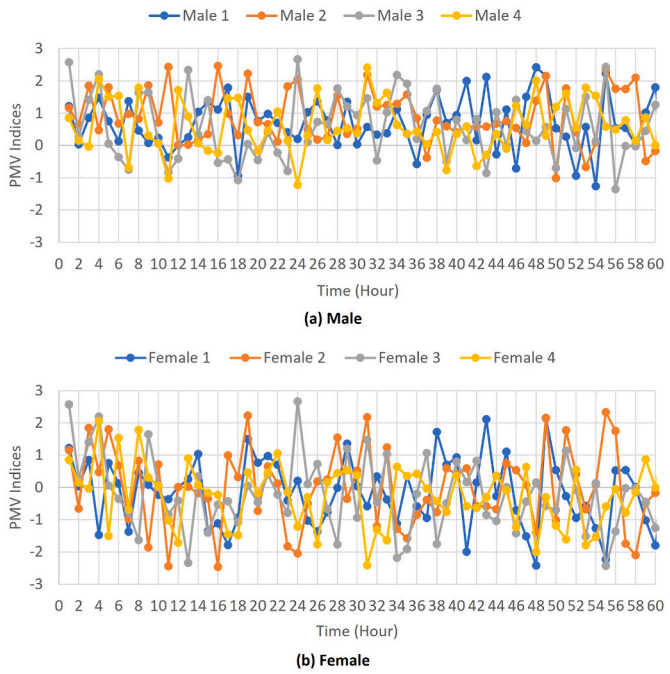


Fig. 7. Stochastic thermal sensation levels (PMV indices) by gender.

the female occupants perceived a slightly cool environment (PMV <0). Similarly, the evaluation of gender-related thermal comfort using the Likert scale revealed a higher thermal discomfort rate for female occupants than males. Some earlier studies have also shown that female occupants are usually 1.5–2 times more likely to be dissatisfied with the indoor environment and layouts than males [7,17]. A study also showed that female occupants' IEQ factor scores were lower than male occupants' scores [53]. Generally, occupants' comfort and thermal sensation levels depend not only on a particular socio-demographic factor or interior layout configuration but also on dynamic indoor environmental factors such as temperature, humidity, and CO_2 levels. Occupants' metabolic heat gains and clothing level are also crucial determinants of the comfort level [7,23,65].

4.3. Reliability study

Generally, indoor data in the spatial layout corresponds to individual occupants' thermal comfort and sensation levels. For a reliability analysis, the data obtained using the statistical and ABM approaches were further studied using ASHRAE 7-point PMV indices. Here, selected occupants (both female and male) from several identical apartments were invited to assess their thermal sensation based on the ASHRAE 7-point scale. The selected occupants were asked to mark anywhere on the scale freely. These are comparable values to those that Fangers' PMV equation [73] attempts to predict. Finally, to evaluate the PMV indices relative to the statistical and ABM outcomes, the male and female PMV indices were assessed, as shown in Fig. 8. The PMV indices showed that the thermal sensation levels for both females and males greatly varied because of environmental factors as well as physical (interior layout-related) and non-physical (socio-demographic-related) variations. Close to the ABM findings, the empirical data also specified that the female occupants perceived a slightly cooler (e.g., PMV ≈ -1) conditions than the male occupants. However, the ABM-predicted PMV indices were highly divergent from the empirical values. Several reasons may explain this observation. First, occupants' actual PMV indices are significantly influenced by housing/building orientations (e.g., north/-south facing windows, doors etc.) [6,50,71]; however, the suggested ABM did not count any housing/building orientation-related factors. Second, the ABM model did not count the variation of occupant

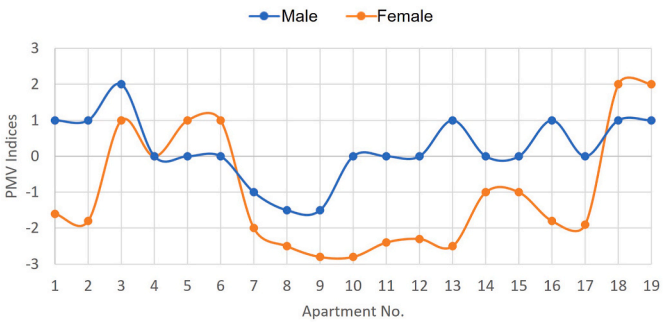


Fig. 8. Thermal sensation levels (PMV indices) by gender for different apartments.

individual clothing levels, which considerably affected the ABM results. Third, appropriate identification of interior layout design or location of furniture/stuff within the ABM-SD model is utterly difficult and can also sometimes lead to inaccurate outcomes [69].

In addition to the above factors, race, attitudes, social norms, and other cultural aspects are significant drivers of the PMV indices [74,75]. Here, "attitude" is an occupant's belief about behavioral influences and their belief that a behavior will ultimately benefit them, while "social norm" relates to the social acceptability or desirability of behaviors.

4.4. Discussion

Occupants' comfort or thermal comfort is the feeling of pleasure that they experience while residing in the thermal atmosphere. In general, occupants evaluate indoor environments based on their needs and desires, which are significantly influenced by their sentimental (subjective) and cognitive (objective) sense. A high degree of similarity between the actual and desired circumstances leads to a great level of thermal comfort for both female and male occupants [71]. However, a discrepancy between desire and building requirements may lead to occupant discomfort or unhappiness. As shown previously [7,17], thermal comfort or comfort temperature and the use of windows are considerably higher in female occupants than male occupants. Moreover, younger occupants (aged ≤25 years) express thermal sensation at a slightly but noticeably higher comfort temperature than older occupants (aged >25 years). These experiences might also be influenced by socio-demographic factors [7]. According to Tan et al. [76], occupant overall comfort may also depend on housing contract programs. Home or building ownership offers a bigger sense of control of the home; the interior layout meets individuals' needs for personal attainment and security. Similarly, landlords or proprietors invest in social assets to build social solidity, cohesion, and interaction among fellow inhabitants. Besides, according to Refs. [6,77], tenants' overall comfort can be classified into four key criteria: layout planning, comfort with the service presented, comfort package offered for the rent, and comfort with the localities or neighborhoods. The present study focused on the occupant-centric thermal comfort level within a particular layout for both landlord and tenant occupants. Additionally, the study attempted to identify the other significant socio-demographic (i.e., gender, age) and physical design factors that play an essential role in occupants' thermal comfort by using both statistical and ABM approaches. At this point, more critical design factors [69] could be included, i.e., occupant destination (e.g., seating position), circulation paths (e.g., walking routes), switch locations for HVAC/lights (i.e., energy spots), room orientation, and furniture positions (e.g., objects), as shown in Fig. 9. This survey study also identified that most of the chosen interior layout patterns were similar to Interior Layout 1 (energy spot inaccessible) shown in Fig. 10. However, the occupants preferred Interior Layouts 2, 3, or 4 (energy spot accessible).

This concern may also elucidate how and why individual occupants

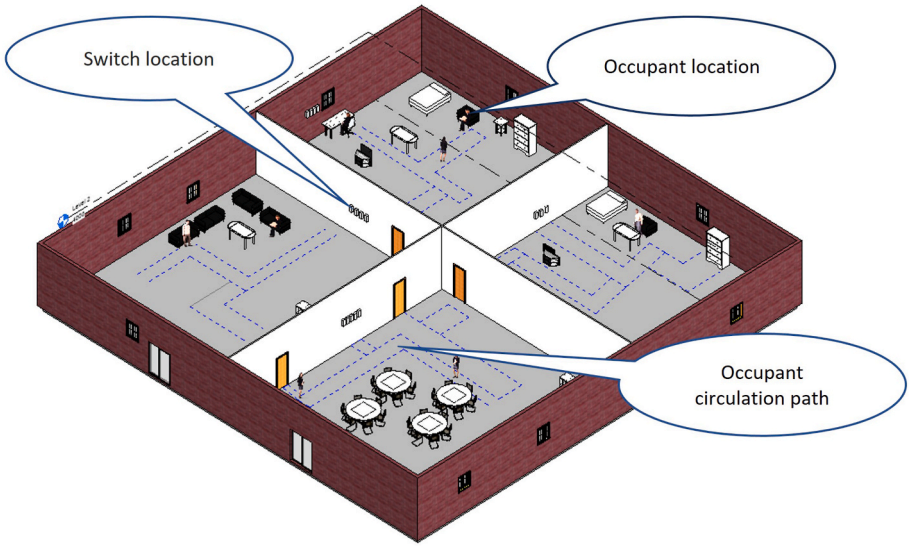


Fig. 9. Occupant seating and circulation paths (adapted from [69]).

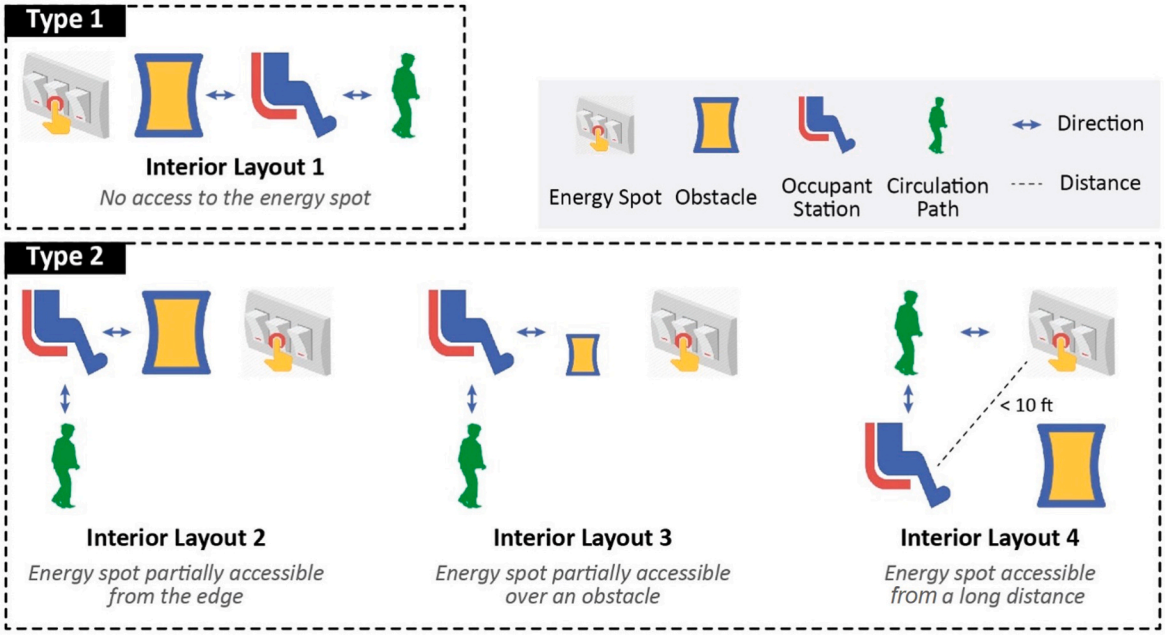


Fig. 10. Existing and probable interior layout pattern within the space (adapted from [69]).

spend more energy for their thermal comfort management. This skill might in turn inform planning for occupancy-based interventions to improve energy saving. Notably, occupants' interior layout deployment is a design task occurring between design development and scheme design in the initial design stage. The occupants' interior layout is an important feature that affects the overall building performance in terms of energy and comfort [6,69,78]. Hence, future studies should comprehensively analyze the currently uncertain influence of occupants' interior layout on comfort and energy performance.

5. Conclusion

By counting both building physical and occupant socio-demographic factors, this study addressed a research gap in the building and occupant comfort literature by investigating the crucial impact of socio-demographical factors on occupants' thermal comfort and sensation.

A semi-structured questionnaire survey was distributed to building occupants in a low-income economy to solicit their opinions on thermal comfort and sensation. The data samples were explored and categorized based on gender, age, and home ownership, followed by a descriptive statistical testing and implementation of the ABM approach. Though the study did not find any substantial influence of home ownership type or age group, both the statistical analysis and ABM indicated that gender exerted a considerable impact on thermal sensation and comfort. Notably, the female occupants were more thermally dissatisfied with the existing interior layout than the male occupants. According to an in-depth analysis, both ABM and the empirical findings showed that the female occupants felt slightly cooler in terms of thermal sensation than the male occupants.

The study elucidates occupants' individual thermal perception and preferences based on specific socio-demographic factors, which will enable stakeholders and policymakers to plan and manage an

appropriate interior layout management for a particular demography. Another contribution of this study is an integrated model (statistical analysis and ABM) that captures the broader aspects of occupants' thermal paradigms in the context of low-income economies. The study also developed an automated integration of the ABM–SD–BIM structure to expedite the study of occupancy-based thermal comfort and sensations, which will be beneficial for architects, engineers, and researchers to improve the existing simulation process. The reliability study also contributes to establishing the ABM model's feasibility and performance.

Nevertheless, occupant perception and comfort are more complex phenomena than other factors related to building indoor performance. It is also important to confine the interior layout plans according to a range of physical parameters to fully determine the occupancy-based thermal and energy performance. If this and other complex issues associated with the configuration of the various interior layouts are not well addressed, it may affect performance throughout the building life cycle. Therefore, further study is required, with more comprehensive data and greater technical sophistication, for building better statistical and ABM models. Nonetheless, this study supports to the global body of knowledge on the occupant socio-demographic factors that influence the relationships between a consistent interior layout and thermal comfort. This may aid the identification of barriers requiring serious interventions to accelerate the adoption of sustainable housing or energy-efficient technologies. The validation approach using real data gathered from building occupants indicated the simulation reliability and the robustness and trustworthiness of the ABM model.

CRediT authorship contribution statement

Mohammad Nyme Uddin: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Minhyun Lee:** Writing – review & editing, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Meng Ni:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.buildenv.2023.110974.

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